

L12 SU Candidate Leg C26 — Cat 6 Substrate-Symplectic UNIVERSAL

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L12 SU Candidate Leg C26 v0.1

Claim

Cat 6 substrate-symplectic substrate-mechanism is UNIVERSAL across substrate-K-type spectrum on D_{IV}^5 via $Sp(2) \cong Spin(5)$ accidental Lie-group isomorphism.

Independent Substrate-Mechanism Support

1. $Spin(5)$ is the double cover of $SO(5)$; $SO(5)$ acts on substrate-K-types $V_{(\lambda_1, \lambda_2)}$
2. $Sp(2) \cong Spin(5)$ accidental isomorphism (standard Lie-group representation theory)
3. $Sp(2)$ substrate-symplectic 2-form ω substrate-canonical antisymmetric form on V_{fund}
4. Substrate-Hamiltonian flow on substrate-K-type spectrum via $Sp(2)$ substrate-symplectic representation theory
5. Substrate-time evolution $\rho_{commit}(\tau) = \exp(-\tau H_B / \hbar_{BST})$ substrate-Hamiltonian flow per $Sp(2)$ substrate-symplectic

Cross-Domain Failure if D_{IV}^5 Generic

If D_{IV}^5 were generic bounded symmetric domain WITHOUT $Sp(2) \cong Spin(5)$ accidental isomorphism, substrate-symplectic substrate-mechanism would NOT exist UNIVERSALLY across substrate-K-types. The accidental isomorphism is specific to substrate-rank=2 + substrate-genus=5 Hermitian symmetric domain.

Strong-Uniqueness Contribution

Adds 1 leg to v1.6 STANDALONE 11 legs $\rightarrow$ 12 if ratified.

Cumulative null-model: $(1/3)^{12} \approx 1.9 \times 10^{-6}$.

Multi-Week RIGOROUS Promotion Path

- substrate-symplectic substrate-Pfaffian per K-type explicit derivation
- substrate-Hamiltonian flow $Sp(2)$ representation theory at substrate-K-type spectrum
- substantive cross-K-type substrate-symplectic UNIVERSAL substantive verification

— Lyra, Thu 2026-06-04 11:23 EDT. L12 SU candidate leg C26 v0.1: Cat 6 substrate-symplectic UNIVERSAL via $Sp(2) \cong Spin(5)$ accidental isomorphism; substantive cross-K-type substrate-mechanism layer; SU 12 if ratified ($1.9e-6$ null-model).